Scout → VECTOR: Modeling Status Brief (Barabási / Vespignani / Gates briefing)

Date: 2026-05-24
Purpose: Give **VECTOR** (manuscript LLM) an accurate picture of **hard-won findings, current implementation, and forward work** so Charles and Alex can brief **László Barabási** and committee member **Alessandro Vespignani** in Boston (~late May 2026).
Author: Scout — Cursor coding agent on the Ivy_Net workspace (pipelines, generative sims, notebooks, HPC). Scout implements and stress-tests; **VECTOR** carries theory and prose; **Alex Gates** shapes sequencing and publication logic.

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1. Advisory context (for VECTOR's voice, not Scout's)

Person	Role
László Barabási	Charles's **official PhD advisor** (networks, Science of Success / Science of Science tradition). Familiar with the **Wang** failure/success template.
Alex Gates	Highly published deputy who ran one of Laszlo's **Science of Success / Science of Science** lines at Northeastern. When Alex took a faculty role at **UVA**, Charles followed to stay close to that intellectual line—with **better day-to-day access** than relying on Laszlo's heavy travel schedule.
Alessandro Vespignani	On Charles's **committee**; expected in the Boston briefing. Natural angles: networked populations, contagion-like diffusion of opportunity/reputation, systemic constraints on "success."
VECTOR	Scholar GPT agent helping Charles turn empirical + generative work into a **manuscript** and advisor-ready narrative.

This document is **not** the oral script. It is **ground truth for VECTOR** so the representation of the project stays aligned with what the code and data actually support.

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2. Who Scout is (one short paragraph)

Scout is the hands-on engineering agent in Charles's Cursor workspace: Army tenure pipelines, college basketball panel builds, generative Tier 1 modules (`tier1_pool_assignment.py`, `538` / `538D` notebooks, parameter sweeps on Rivanna/Mac). Scout does **not**

set dissertation theory; Scout makes the **minimal model architecture** runnable, documented, and honest about what is proven vs in progress.

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3. Scientific background (Wang + inverted-U — Laszlo knows this frame)

The dissertation line follows the **Yin–Wang logic**: build a **minimal generative story**, show when a stylized fact **emerges** or **fails**, then refine mechanisms—**not** treat a quadratic reduced form as the primitive.

Stylized fact (empirically replicated): an **inverted-U** between **local pool quality** (leave-one-out mean peer performance L_Q) and **global advancement** Y (Army promotion; NCAA → NBA draft).

Empirical ladder (on *realized* pools in each domain):

1. Descriptive bins
2. Transparent quadratic / LPM in L + ability controls
3. Logit/probit, report turning point L^*
4. Robustness

That ladder is **operational in basketball** (`538_alex_tier1_model_and_fit.ipynb`) and was the path to **replicating the Army result** in sports—Charles's strongest "it's not just one dataset" evidence.

Conceptual spine (cross-domain):

> *Advancement under **scarce global selection** when people sit in **local, non-random pools**. Stronger peers can raise **development and signal** but lower **distinctiveness** through **comparison, crowding, and congestion**. Net effects can **rise then fall** in pool quality.*

That framing fits Barabási's networks (assortative structure, local neighborhoods) and Vespignani's systems view (finite "slots," interaction-driven outcomes).

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4. What we are building (the real center — not one tuning knob)

4.1 Minimal model first, rich variables second

The goal is a **domain-agnostic minimal Tier 1 model** with **modular pieces** that can later be **decomposed** into interpretable constructs:

Construct (language for the brief)	Role in the minimal stack	Code / data hooks
Own ability / performance A_i	Baseline human capital	Panel perf; generative draws
Local pool quality L_Q	LOO mean peer ability — "who surrounds you"	`poolq_loo`, congestion_quality
Crowding / congestion L_C	Viable-peer density, comparison cost	`pool_c_loo`, smooth viability σ(γ(A−θ))
Pool dispersion	Peer SD — distinction vs compression	`peer_perf_sd_loo`, 538 CELL 4B/4C
Development / opportunity (implicit)	Upside from strong environment (not yet separate equation)	Theory + future decomposition
Global selection capacity Λ	Top-K, top 10%, draft slots	`N_SELECTED`, draft indicator
Pre-sorting / assignment	How people land in pools (overlap, assortativity)	Soft assign to T_j; forensics vs sort-and-chop

Important for VECTOR: The research is **not** "find the right τ." Assignment temperature (τ) is **one implementation dial** for how tightly people match pool targets T_j. The scientific object is **how multiple local indices enter promotion rules in opposing directions**—quality up, congestion down, dispersion ambiguous—on top of **non-random pool formation**.

4.2 Two layers (do not merge in the manuscript)

Layer	Question	Status
Empirical	Given **real** rosters (team-season, unit, lab), what is the shape of Y vs L?	**Strong** — Army + basketball inverted-U, Wang ladder, exploratory mean×SD
Generative	What **rules** (assignment + selection) produce overlap and an inverted-U **without** assuming L^2 in Y?	**Active** — playground + sweeps; mechanism U not yet demonstrated

Alex's post–Paper Directions 10 push: **explain the curve** from sorting + local comparison, not **fit the curve** first.

4.3 Alex's one-night notebook (`539_alex_model.ipynb`) — useful sketch, not the spine

Alex produced a **quick proof-of-concept** (assortative sort-and-chop, smooth viability, eval noise, global top-10%). It helped the team **see** congestion in one place. It is **not** the dissertation architecture and should **not** dominate the Barabási brief.

VECTOR should treat 539 as: "Alex's sandbox."
VECTOR should treat 538 + `tier1_pool_assignment.py` as: "Charles's modular minimal model lab" (soft pools, swappable L in the

score, top-K or quantile selection later).

Internal engineering once overweighted **matching 539's sort-and-chop** via τ calibration. That was a **development detour**, not the scientific headline. The headline is **multi-term local environment → global outcome**.

4.4 Cross-domain program (why Laszlo and Alessandro should care)

Same **logical skeleton** across **Army, academia, sports**:

- **Local** interaction field (peers, coauthors, teammates)
- **Global** scarce promotion (general, tenure, draft)
- **Non-random** placement into local environments

Basketball is the **cleanest measurement lab** (global draft vs local minutes/roles). Army is the **anchor replication**. Academia is on the roadmap with the same Tier 1 objects. The generative layer is deliberately **not** "copy empirical Duke's roster means"—it uses **portable rules** (draw T_j, soft match, score = A ± weighted local L terms).

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5. Where we are now (findings + implementation — honest)

5.1 Findings VECTOR can assert

1. **Inverted-U replicated** in college basketball on realized pools, parallel to Army.
2. **Real pools overlap** massively on the performance axis (530 forensics); legacy **sort-and-chop** simulation does **not** (disjoint slices, coverage ≈ 1).
3. **Generative direction is live:** soft assignment to target means T_j reproduces **overlap-rich** rosters when calibrated to **real-data forensics** (median within-roster SD ~0.8 z-scale in basketball).
4. **Mechanism variables are wired modularly:** quality L_Q, crowding share L_C, smooth congestion, gap-to-pool score $w\cdot(A - L_Q)$, optional preferential attachment—each can be turned on/off in CELL 10.
5. **Exploratory PD11-B:** mean peer quality × peer SD vs draft (538 CELL 4B/4C) tests whether **dispersion** belongs beside **level**.

5.2 Work in progress (VECTOR should phrase as "building," not "done")

1. **Generative inverted-U:** with **ability-only** selection, Plot B vs L_Q rises (sorting + merit)—expected **null floor**. **Congestion/crowding in the selection score** is the next step to seek a **peak and right-tail drop**.
2. **Unified story figure:** one slide-quality generative panel that mirrors Wang "emergence" without a quadratic outcome spec.
3. **Parameter sweeps (HPC):** `faithful_538` grid explores assignment + selection combos; queued on Rivanna—feeds **identifiability**, not the oral centerpiece.
4. **Manuscript:** VECTOR + Alex outline ahead of a single **Barabási-ready** mechanism cartoon (pools → local indices → global Y).

5.3 What to downplay in the Boston room

- Long explanations of **τ = 0.65 vs 0.11** or "539 assignment calibration."
- Notebook numbering (`537` frozen, `538` forward) unless someone asks where code lives.

5.4 What to emphasize

- **Replication + mechanism ambition** (Wang template).

- **Multiple local forces in concert** (quality, congestion, crowding, future development term).
- **Minimal portable model** with **decomposable modules** for cross-domain Science of Success.
- **Charles's structural choice** to follow Alex at UVA while Laszlo remains formal advisor—intellectual continuity with the Northeastern line.

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6. Suggested narrative beats for VECTOR (Barabási + Vespignani + Gates)

1. **Empirical win:** inverted-U in two domains on **real** local pools; L^* reported via transparent ladder.
2. **Theoretical move:** congestion/crowding as **consequence of correlated pools + local comparison**, not only a second covariate.
3. **Generative program:** replace unrealistic sort-and-chop with **overlapping pools**; build **selection scores** that combine A with L_Q, L_C, dispersion—tunable weights, domain-agnostic.
4. **Network hook for Laszlo:** assortativity creates **correlated neighborhoods**; advancement is a **global filter** on locally defined standing.
5. **Systems hook for Alessandro:** finite promotion capacity + interaction structure → **nonlinear macro pattern** (inverted-U) from micro rules.
6. **Near-term ask:** publish **empirical cross-domain** while generative mechanism figure matures, or hold for one generative "U emerges" demo—advisor judgment call.
7. **Gratitude / lineage:** Science of Success framing; Alex as day-to-day methodological partner; Laszlo as dissertation anchor.

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7. Forward work (6–12 month vector for VECTOR)

Priority	Task
1	Turn on **congestion/crowding** in generative selection; test inverted-U in L_Q bins without $Y \sim L^2$
2	Name and optionally separate **development upside** vs **comparison cost** in the score (theory-led decomposition)
3	One **cross-domain schematic** (Army
4	Finish **538 empirical** robustness (CELL 7+); keep generative in `538D` until stable
5	Manuscript section: **empirical replication** first; **generative mechanism** second—Wang ordering

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8. Key files (VECTOR citations)

Artifact	Path
Advisor spine	`Alex_Tier1_Sequential_Model_Outline.md`
Design note (pools + 530 forensics)	`sports/documents/Tier1_Presorting_Design_Note.md`
VECTOR theory	`Vector_Questions_and_Modeling_Thoughts.md`, `Vector_to_Scout_Tier1_Modeling_Direction.md`
Alex simulation memo (PD10)	`2026_0507_Alex_Gates_Post_Meeting_Simulation_Memo.md`
Empirical ladder	`sports/538_alex_tier1_model_and_fit.ipynb`
Generative lab	`sports/538D_development.ipynb`, `tier1_pool_assignment.py`, `tier1_sim_config.py`
Real-data forensics	`sports/530_sports_pipeline.ipynb`
Alex one-night POC (optional)	`sports/539_alex_model.ipynb`
Legacy sim (comparison only)	`537` notebooks, `sim_config.py`

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9. One-paragraph summary VECTOR can paste

Charles Levine's dissertation (advisor **László Barabási**, day-to-day methodological partner **Alex Gates** at UVA, committee including **Alessandro Vespignani**) studies **advancement under constrained distinction**: finite global selection when individuals occupy **non-random local pools**. Empirically, he replicated the Army **inverted-U** between leave-one-out pool quality and promotion in **college basketball → NBA draft**, using a **Wang-ordered** ladder on realized rosters—not a quadratic-first story. The forward program is a **minimal cross-domain generative model** whose modules—**ability, pool quality, crowding/congestion, dispersion, assignment rules, global capacity**—can be switched and later **decomposed** into development vs comparison narratives. Implementation is in **538** (empirical + generative playground); **530** forensics ground pool realism; Alex's **539** notebook is a one-night POC, not the architecture. **Scout** maintains the code path; **VECTOR** should represent **findings as solid**, **mechanism as actively under construction**, and **τ** as a minor assignment dial—not the scientific center of gravity.

Scout brief ends. VECTOR: reframe any draft that over-indexes on τ or 539; lead with multi-variable local environment, cross-domain minimal model, and replicated inverted-U.